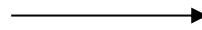


Homogeneous Coordinates and Matrix Representation

Representing All transformation
(Translations, rotation, scale) as a
single operation



Matrix multiplication

Canonical forms

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Translation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Scaling

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Rotation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & Sh_x & 0 \\ Sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Shearing /shear

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

X Mirror / Reflect

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Y Mirror / Reflect

Properties of translations

1. $T(0,0) = I \rightarrow$ The Identity Translation

Meaning – Translating by zero distance gives you the identity matrix (no movement)

$$T(0,0) \cdot P(x,y,1) = P(x,y,1)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

2. $T(s_x, s_y) \cdot T(t_x, t_y) = T(s_x + t_x, s_y + t_y) \rightarrow$ Combining Translations

Meaning – Applying two translations consecutively is equivalent to applying on combined translation.

Eg ; Let's say we want to translate a point first by $T_1(2, 3)$ and then by $T_2(1, -1)$

$$T_{combined} = T_2 \times T_1 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

3. $T(s_x, s_y) \cdot T(t_x, t_y) = T(t_x, t_y) \cdot T(s_x, s_y) \rightarrow$ Commutativity

Meaning : The order of translation doesn't matter !

4. $T^{-1}(s_x, s_y) = T(-s_x, -s_y) \rightarrow$ Inverse translation

Meaning :- To undo (inverse) the translations. Just translate in the opposite direction.

$$T^{-1}(s_x, s_y) = T(-s_x, -s_y) = \begin{bmatrix} 1 & 0 & -s_x \\ 0 & 1 & -s_y \\ 0 & 0 & 1 \end{bmatrix}$$

Properties of scaling

1. Identity Property

Scaling by (1, 1) is the identity operation: $S(1, 1) \times S(sx, sy) = S(sx, sy)$

I $\times$ **S**:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2. Commutative Property

Scaling operations are commutative: $S(sx_1, sy_1) \times S(sx_2, sy_2) = S(sx_2, sy_2) \times S(sx_1, sy_1)$

S₁ $\times$ **S**₂:

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

S₂ $\times$ **S**₁:

$$\begin{bmatrix} 4 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3. Inverse Property

The inverse of scaling is scaling by reciprocals: $S^{-1}(sx, sy) = S(1/sx, 1/sy)$

Original scaling: S(2, 4)

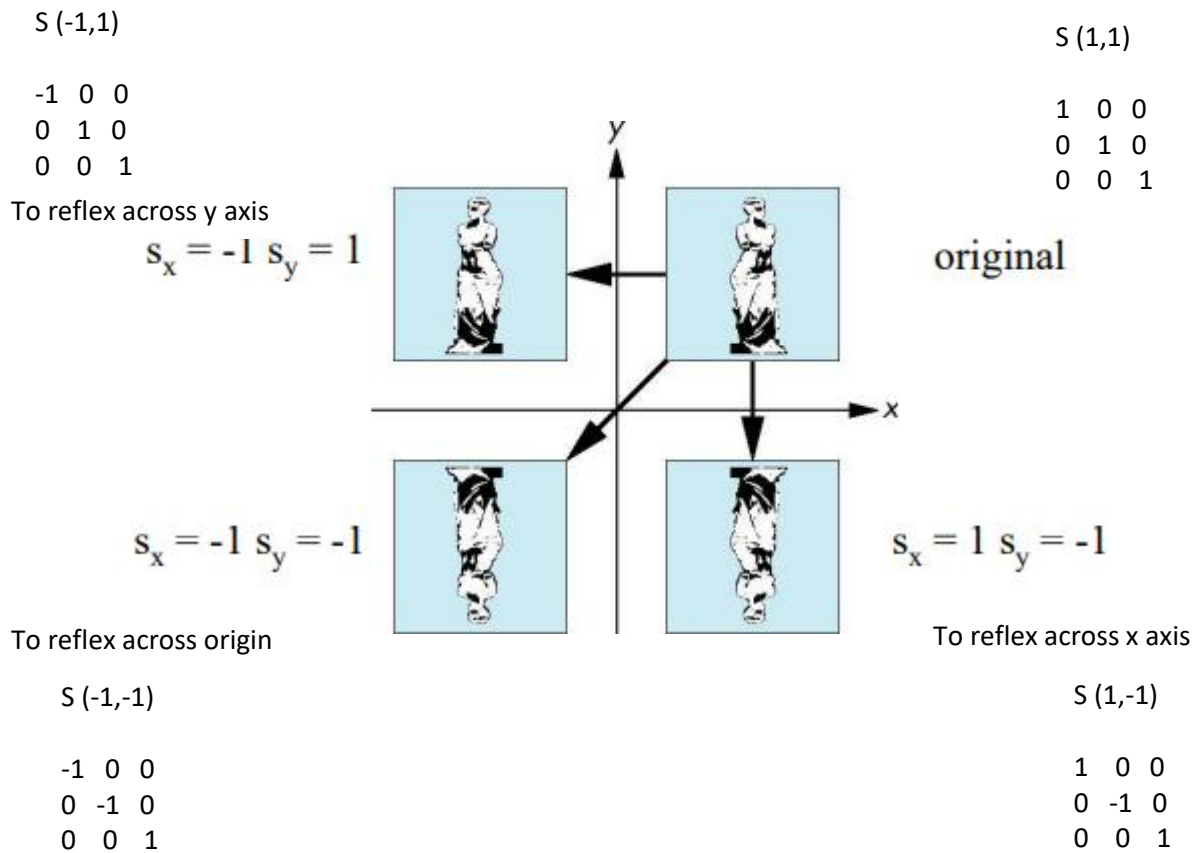
$$S = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Inverse scaling: S(0.5, 0.25)

$$S^{-1} = \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Properties of Reflection

Reflection can be represented as a scaling transformation with one or more negative scale factors.



Properties of Rotation

1. $\mathbf{R}^{-1}(\theta) = \mathbf{R}(-\theta) = \mathbf{R}^T \rightarrow$ Inverse Property

$\mathbf{R}^T$ is the transpose. Transpose means flipping the rows and columns

Example: Rotation by 60°

Original rotation: $\mathbf{R}(60^\circ)$

$$\mathbf{R} = \begin{bmatrix} \cos 60^\circ & -\sin 60^\circ & 0 \\ \sin 60^\circ & \cos 60^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Inverse rotation: $\mathbf{R}(-60^\circ) = \mathbf{R}^T$

$$\mathbf{R}^{-1} = \begin{bmatrix} \cos(-60^\circ) & -\sin(-60^\circ) & 0 \\ \sin(-60^\circ) & \cos(-60^\circ) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.866 & 0 \\ -0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{R}^T$$

2. $\mathbf{R}(\theta) \times \mathbf{R}(\phi) = \mathbf{R}(\theta + \phi) \rightarrow$ Composition Property

Example: Rotate by 30° then by 45° = Rotate by 75°

$\mathbf{R}(30^\circ) \times \mathbf{R}(45^\circ)$:

$$\begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.707 & -0.707 & 0 \\ 0.707 & 0.707 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.259 & -0.966 & 0 \\ 0.966 & 0.259 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\mathbf{R}(75^\circ)$:

$$\begin{bmatrix} \cos 75^\circ & -\sin 75^\circ & 0 \\ \sin 75^\circ & \cos 75^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.259 & -0.966 & 0 \\ 0.966 & 0.259 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3. $\det(\mathbf{R}) = 1 \rightarrow$ Determinant Property

Meaning: Rotations preserve area, orientation and angles

Example: For any rotation $\mathbf{R}(\theta)$

$$\det(\mathbf{R}) = \cos^2 \theta + \sin^2 \theta = 1$$

For $\mathbf{R}(60^\circ)$:

$$\det \begin{bmatrix} 0.5 & -0.866 \\ 0.866 & 0.5 \end{bmatrix} = (0.5)(0.5) - (-0.866)(0.866) = 0.25 + 0.75 = 1$$

4. $\mathbf{R}^T \times \mathbf{R} = \mathbf{R} \times \mathbf{R}^T = \mathbf{I} \rightarrow$ Orthogonal Property

Meaning : Calling a rotation matrix "**orthogonal**" is just the mathematical way of guaranteeing that the transformation it represents is a **pure rotation**—nothing gets distorted, and reversing it is as simple as doing the opposite turn.

Example: For $R(60^\circ)$

$$\mathbf{R}^T \times \mathbf{R} = \begin{bmatrix} 0.5 & 0.866 & 0 \\ -0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Task 1 : Understanding transformation Properties – Use Matrixes when answering

Submission : Write answers on a paper and upload the scanned images in a pdf format.

1. What is the result of applying the translation $T(0, 0)$ to any point $P(x, y)$? What is the matrix representation of this translation?
2. If you translate a point first by $T_1(2, 3)$ and then by $T_2(1, -1)$, what is the single, combined translation matrix that achieves the same result?
3. Is the order of applying translations important? Demonstrate with $T(2, 3)$ and $T(1, -1)$
4. What is the inverse of the translation $T(5, -2)$, and what is its geometric meaning?
5. What is the identity operation for scaling, and what is its matrix?
6. Show that scaling is commutative by calculating $S(2, 3) \cdot S(4, 0.5)$ and $S(4, 0.5) \cdot S(2, 3)$.
7. What is the inverse of the scaling transformation $S(4, 5)$? Show how applying the inverse after the original scaling returns a point to its original location.
8. What scaling values would you use to create a reflection across the y-axis? Write its matrix.
9. What is the matrix for reflecting a point across the origin?
10. What are the two equivalent ways to find the inverse of a rotation matrix $R(\theta)$?
11. If you rotate a point by 30° and then by 45° , what is the single equivalent rotation?
12. What is the determinant of any standard 2D rotation matrix $R(\theta)$, and what does this value tell us about the transformation?
13. What is the defining property of an orthogonal matrix, and how does it manifest for a rotation matrix $R(\theta)$?

Task 2 : Applying Properties of transformation

Submission : Write answers on a paper and upload the scanned images in a pdf format.

1. A point $P(2, 3)$ is first scaled by $S(2, 0.5)$ and then translated by $T(4, -1)$. Calculate the final position of the point by finding the combined transformation.
2. Find the inverse of the transformation matrix M that represents a translation $T(3, 2)$ followed by a scaling $S(2, 2)$. Show that $M^{-1} \times M = I$.
3. Find the transformation matrix to rotate a point by 90° counterclockwise about the point $C(2, 3)$. (This requires translating C to origin, rotating, then translating back)
4. Find the matrix for reflecting a point across the vertical line $x = 3$. (Translate line to y-axis, reflect, translate back)
5. Verify that for $R(60^\circ)$, both $R^{-1} = R(-60^\circ)$ and $R^{-1} = R^T$ hold true. Also verify the orthogonal property $R^T \times R = I$.